

Revision Notes for Class 12 Mathematics

Chapter 8 – Application of Integrals

Definite Integration

1. Definition:

If $F(x)$ is an antiderivative of $f(x)$, then $F(b) - F(a)$ is known as the definite integral of $f(x)$ from a to b , such that the variable x , takes any two independent values say a and b .

This is also denoted as $\int_a^b f(x)dx$.

Thus $\int_a^b f(x)dx = F(b) - F(a)$, The numbers a and b are called the limits of integration; a is the lower limit and b is the upper limit.

2. Properties of Definite Integrals:

$$\text{I. } \int_a^b f(x)dx = - \int_b^a f(x)dx$$

$$\text{II. } \int_a^b f(x)dx = \int_b^a f(y)dy$$

$$\text{III. } \int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx, \text{ where } c \text{ may or may not lie between } a \text{ and } b.$$

$$\text{IV. } \int_0^a f(x)dx = \int_0^a f(a-x)dx$$

$$V. \int_a^b f(x)dx = \int_a^b f(a+b-x)dx$$

Note:

$$a. \int_0^a \frac{f(x)}{f(x)+f(a-x)}dx = \frac{a}{2}$$

$$b. \int_a^b \frac{f(x)}{f(x)+f(a+b-x)}dx = \frac{b-a}{2}$$

$$VI. \int_0^{2a} f(x)dx = \int_0^a f(x)dx + \int_0^a f(2a-x)dx$$

$$= \begin{cases} 0 & \text{if } f(2a-x) = -f(x) \\ 2 \int_0^a f(x)dx & \text{if } f(2a-x) = f(x) \end{cases}$$

$$VII. \int_{-a}^a f(x)dx = \begin{cases} 2 \int_0^a f(x)dx & \text{if } f(-x) = f(x) \text{ i.e. } f(x) \text{ is even} \\ 0 & \text{if } f(-x) = -f(x) \text{ i.e. } f(x) \text{ is odd} \end{cases}$$

VIII. If $f(x)$ is a periodic function of period T (Text translation failed), i.e. $f(a+x) = f(x)$, then

$$a. \int_0^{na} f(x)dx = n \int_0^a f(x)dx$$

$$b. \int_0^{na} f(x)dx = (n-1) \int_0^a f(x)dx$$

c. $\int_{na}^{b+na} f(x)dx = \int_0^b f(x)dx$, where $b \in \mathbb{R}$

d. $\int_b^{b+a} f(x)dx$ independent of b .

e. $\int_b^{b+na} f(x)dx = n \int_0^a f(x)dx$, where $n \in \mathbb{N}$

IX. If $f(x) \geq 0$ on the interval $[a, b]$, then $\int_a^b f(x)dx \geq 0$.

X. If $f(x) \leq g(x)$ on the interval $[a, b]$, then $\int_a^b f(x)dx \leq \int_a^b g(x)dx$

XI. $\left| \int_a^b f(x)dx \right| \leq \int_a^b |f(x)|dx$

XII. If $f(x)$ is continuous on $[a, b]$, m is the least and M is the greatest value of $f(x)$ on $[a, b]$, then

$$m(b-a) \leq \int_a^b f(x)dx \leq M(b-a)$$

XIII. For any two functions $f(x)$ and $g(x)$, integral on the interval $[a, b]$, the Schwarz-Bunyakovsky inequality holds

$$\left| \int_a^b f(x).g(x)dx \right| \leq \sqrt{\int_a^b f^2(x)dx} \cdot \sqrt{\int_a^b g^2(x)dx}$$

XIV. If a function $f(x)$ is continuous on the interval $[a, b]$, then there exists a point $c \in (a, b)$ such that

$$\int_a^b f(x) dx = f(c)(b-a), \text{ where } a < c < b.$$

3. Differentiation Under Integral Sign:

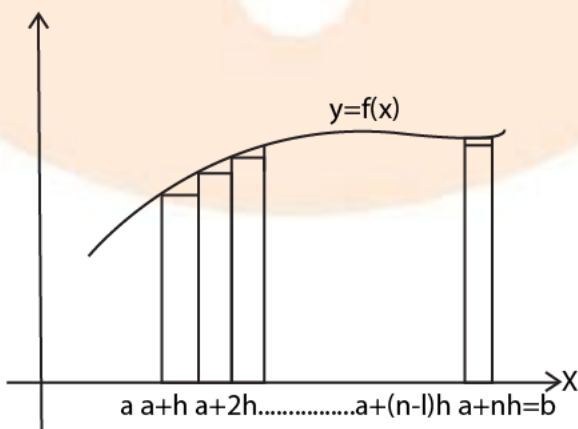
Newton Leibnitz's Theorem:

Given that x has two differentiable functions, $g(x)$ and $h(x)$, where $x \in [a, b]$ and f is continuous in that interval, then

$$\frac{d}{dx} \left[\int_{g(x)}^{h(x)} f(t) dt \right] = \frac{d}{dx} [h(x)] \cdot f[h(x)] - \frac{d}{dx} [g(x)] \cdot f[g(x)]$$

4. Definite Integral as a Limit of Sum:

Let $f(x)$ be a continuous real valued function defined on the closed interval $[a, b]$ which is divided into n parts as shown in figure.



The point of division on x -axis are

$$a, a+h, a+2h, \dots, a+(n-1)h, a+nh, \text{ where } \frac{b-a}{n} = h.$$

Let S_n denotes the area of these n rectangles.

$$\text{Then, } S_n = hf(a) + hf(a+h) + hf(a+2h) + \dots + hf(a+(n-1)h)$$

Clearly, S_n is area very close to the area of the region bounded by

Curve $y = f(x)$, x -axis and the ordinates $x = a$, $x = b$.

$$\text{Hence } \int_a^b f(x)dx = \lim_{n \rightarrow \infty} S_n$$

$$\int_a^b f(x)dx = \lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} hf(a+rh) = \lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} \left(\frac{b-a}{n} \right) f \left(a + \frac{(b-a)r}{n} \right)$$

Note:

(a) We can also write

$$S_n = hf(a+h) + hf(a+2h) + \dots + hf(a+nh)$$

$$\int_a^b f(x)dx = \lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{b-a}{n} \right) f \left(a + \left(\frac{b-a}{n} \right) r \right)$$

$$(b) \text{ If } a = 0, b = 1, \int_0^1 f(x)dx = \lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} \frac{1}{n} f \left(\frac{r}{n} \right)$$

Steps to Express the Limit of Sum as Definite Integral

Step 1. Replace $\frac{r}{n}$ by x , $\frac{1}{n}$ by dx and $\lim_{n \rightarrow \infty} \sum$ by $\int$

Step 2. Evaluate $\lim_{n \rightarrow \infty} \left(\frac{r}{n}\right)$ by putting least and greatest values of r as lower and upper limits respectively.

For example, $\lim_{n \rightarrow \infty} \sum_{r=1}^{pn} \frac{1}{n} f\left(\frac{r}{n}\right) = \int_0^p f(x) dx$

$$\lim_{n \rightarrow \infty} \left(\frac{r}{n}\right) \Big|_{r=1} = 0, \quad \lim_{n \rightarrow \infty} \left(\frac{r}{n}\right) \Big|_{r=np} = p$$

5. Reduction Formulae in Definite Integrals

I. If $I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx$, then show that $I_n = \left(\frac{n-1}{n}\right) I_{n-2}$

Proof: $I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx$

$$I_n = \left[-\sin^{n-1} x \cos x \right]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} (n-1) \sin^{n-2} x \cdot \cos^2 x dx$$

$$= (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \cdot (1 - \sin^2 x) dx$$

$$= (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x dx - (n-1) \int_0^{\frac{\pi}{2}} \sin^n x dx$$

$$I_n + (n-1)I_n = (n-1)I_{n-2}$$

$$I_n = \left(\frac{n-1}{n} \right) I_{n-2}$$

Note:

$$(a) \int_0^{\frac{\pi}{2}} \sin^n x dx = \int_0^{\frac{\pi}{2}} \cos^n x dx$$

$$(b) I_n = \left(\frac{n-1}{n} \right) \left(\frac{n-3}{n-2} \right) \left(\frac{n-5}{n-4} \right) \dots I_0 \text{ or } I_1 \text{ according as } n \text{ is even or odd, } I_0 = \frac{\pi}{2}, I_1 = 1$$

$$\text{Hence } I_n = \begin{cases} \left(\frac{n-1}{n} \right) \left(\frac{n-3}{n-2} \right) \left(\frac{n-5}{n-4} \right) \dots \left(\frac{1}{2} \right) \cdot \frac{\pi}{2} & \text{if } n \text{ is even} \\ \left(\frac{n-1}{n} \right) \left(\frac{n-3}{n-2} \right) \left(\frac{n-5}{n-4} \right) \dots \left(\frac{2}{3} \right) \cdot 1 & \text{if } n \text{ is odd} \end{cases}$$

$$\text{II. If } I_n = \int_0^{\frac{\pi}{4}} \tan^n x dx, \text{ then show that } I_n + I_{n-2} = \frac{1}{n-1}$$

$$\text{Proof: } I_n = \int_0^{\frac{\pi}{4}} (\tan x)^{n-2} \cdot \tan^2 x dx$$

$$= \int_0^{\frac{\pi}{4}} (\tan x)^{n-2} (\sec^2 x - 1) dx$$

$$= \int_0^{\frac{\pi}{4}} (\tan x)^{n-2} \sec^2 x dx - \int_0^{\frac{\pi}{4}} (\tan x)^{n-2} dx$$

$$= \left[\frac{(\tan x)^{n-1}}{n-1} \right]_0^{\frac{\pi}{4}} - I_{n-2}$$

$$I_n = \frac{1}{n-1} - I_{n-2}$$

$$I_n + I_{n-2} = \frac{1}{n-1}$$

III. If $I_{m,n} = \int_0^{\frac{\pi}{2}} \sin^m x \cdot \cos^n x dx$, then show that $I_{m,n} = \frac{m-1}{m+n} I_{m-2,n}$

Proof: $I_{m,n} = \int_0^{\frac{\pi}{2}} \sin^{m-1} x (\sin x \cos^n x) dx$

$$= \left[-\frac{\sin^{m-1} x \cdot \cos^{n+1} x}{n+1} \right]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \frac{\cos^{n+1}}{n+1} (m-1) \sin^{m-2} x \cos x dx$$

$$= \left(\frac{m-1}{n+1} \right) \int_0^{\frac{\pi}{2}} \sin^{m-2} x \cdot \cos^n x \cdot \cos^2 x dx$$

$$= \left(\frac{m-1}{n+1} \right) \int_0^{\frac{\pi}{2}} (\sin^{m-2} x \cdot \cos^n x - \sin^m x \cdot \cos^n x) dx$$

$$= \left(\frac{m-1}{n+1} \right) I_{m-2,n} - \left(\frac{m-1}{n+1} \right) I_{m,n}$$

$$\Rightarrow \left(1 + \frac{m-1}{n+1}\right) I_{m,n} = \left(\frac{m-1}{n+1}\right) I_{m-2,n}$$

$$I_{m,n} = \left(\frac{m-1}{m+n}\right) I_{m-2,n}$$

Note:

(a) $I_{m,n} = \left(\frac{m-1}{m+n}\right) \left(\frac{m-3}{m+n-2}\right) \left(\frac{m-5}{m+n-4}\right) \dots I_{0,n}$ or $I_{1,n}$ according as m is even or odd.

$$I_{0,n} = \int_0^{\frac{\pi}{2}} \cos^n x dx \text{ and } I_{1,n} = \int_0^{\frac{\pi}{2}} \sin x \cdot \cos^n x dx = \frac{1}{n+1}$$

(b) Walli's Formula

$$I_{m,n} = \begin{cases} \frac{(m-1)(m-3)(m-5)\dots(n-1)(n-3)(n-5)\dots}{(m+n)(m+n-2)(m+n-4)\dots} & \text{when both } m, n \text{ are even} \\ \frac{(m-1)(m-3)(m-5)\dots(n-1)(n-3)(n-5)\dots}{(m+n)(m+n-2)(m+n-4)\dots} & \text{otherwise} \end{cases}$$

Area Under the Curves

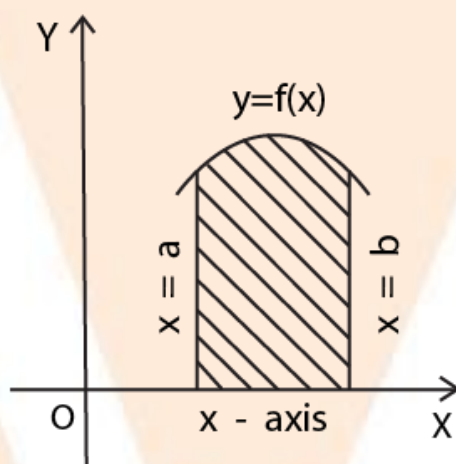
(a) Area of Plane Regions:

1. The area bounded by the curve $y = f(x)$, x -axis and the ordinates $x = a$ and $x = b$ (where $b > a$) is given by

$$A = \int_a^b |y| dx = \int_a^b |f(x)| dx$$

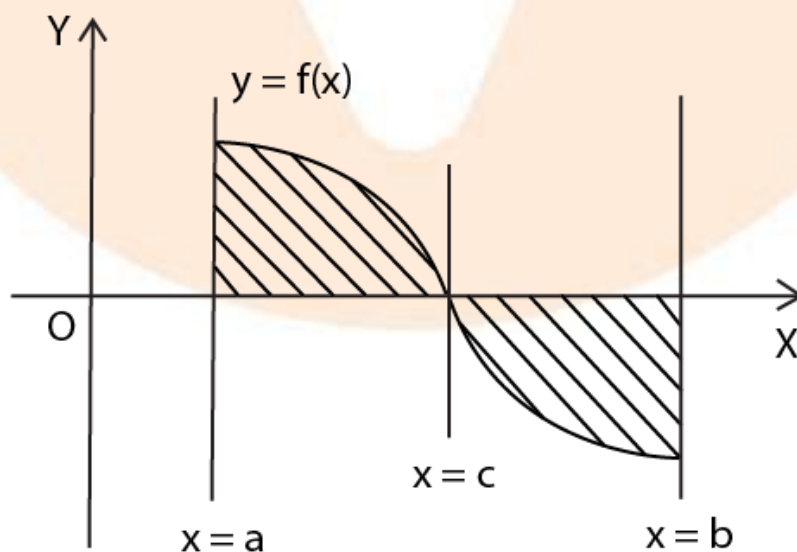
(i) If $f(x) > 0 \forall x \in [a, b]$

Then $A = \int_a^b f(x) dx$



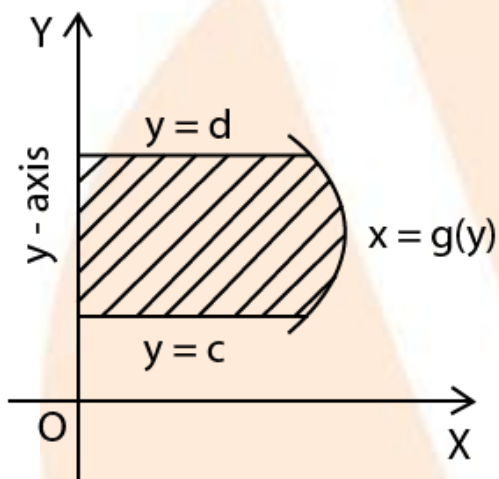
(ii) If $f(x) > 0 \forall x \in [a, c]$ & $< 0 \forall x \in (c, b]$

Then $A = \left| \int_a^c y dx \right| + \left| \int_c^b y dx \right| = \int_a^c f(x) dx - \int_c^b f(x) dx$ where c is a point in between a and b .



II. The area bounded by the curve $x = g(y)$, y -axis and the abscissae $y = c$ and $y = d$ (where $d > c$) is given by

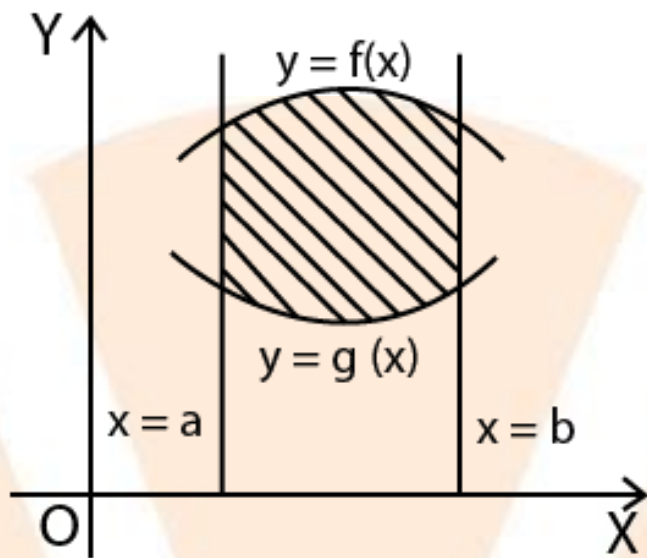
$$A = \int_c^d |x| dy = \int_c^d |g(y)| dy$$



III. If we have two curves $y = f(x)$ and $y = g(x)$, such that $y = f(x)$ lies above the curve $y = g(x)$ then the area bounded between them and the ordinates $x = a$ and $x = b$ ($b > a$), is given by

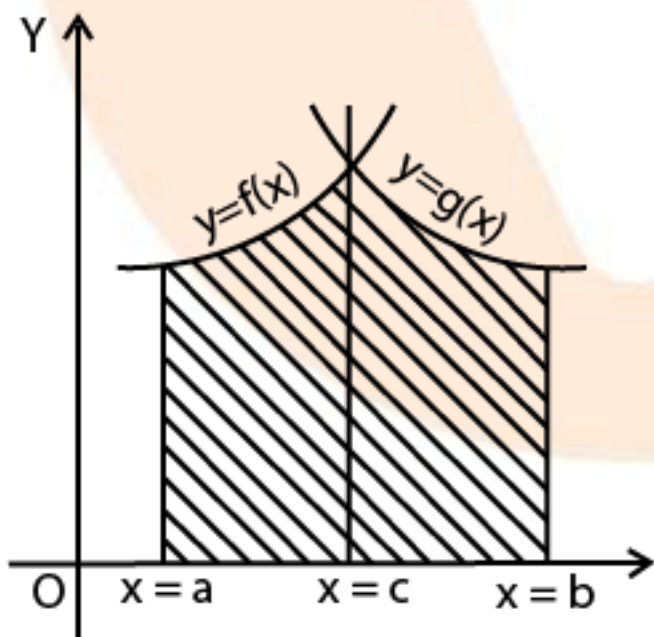
$$A = \int_a^b f(x) dx - \int_a^b g(x) dx$$

i.e. upper curve area – lower curve area .



IV. The area bounded by the curves $y = f(x)$ and $y = g(x)$ between the ordinates $x = a$ and $x = b$ is given by

$$A = \int_a^c f(x)dx + \int_c^b g(x)dx, \text{ where } x = c \text{ is the point of intersection of the two curves.}$$



V. Curve Tracing

It is necessary to have a rough sketch of the desired piece in order to locate the area enclosed by many curves. The following steps are very useful in tracing a cartesian curve $f(x,y) = 0$.

Step 1: Symmetry

- (i) If all the powers of y in the equation of the given curve are even then the curve is symmetrical about x -axis.
- (ii) If all the powers of x in the equation of the given curve are even then the curve is symmetrical about y -axis.
- (iii) If the equation of the given curve remains unchanged on interchanging x and y , then the curve is symmetrical about the line $y = x$.
- (iv) If the equation of the given curve remains unchanged when x and y are replaced by $-x$ and $-y$ respectively, then the curve is symmetrical in opposite quadrants.

Step 2: Origin

If the algebraic curve's equation contains no constant term, the curve passes through the origin.

The tangents at the origin are then calculated by equating the lowest degree terms in the equation of the specified algebraic curve to zero.

For example, the curve $y^3 = x^3 + axy$ passes through the origin and the tangents at the origin are given by $axy = 0$ i.e. $x = 0$ and $y = 0$.

Step 3: Intersection with the Co-ordinates Axes

- (i) Find out the corresponding values of x by putting $y = 0$ in the equation of the given curve, to estimate the points of intersection of the curve with x -axis

(ii) Find out the corresponding values of y by putting $x = 0$ in the equation of the given curve, to estimate the points of intersection of the curve with y -axis

Step 4: Asymptotes

Find out the asymptotes of the curve.

(i) The vertical asymptotes of the given algebraic curve, or asymptotes parallel to the y -axis, are derived by equating the coefficient of the highest power of y in the equation of the supplied curve to zero.

(ii) The horizontal asymptotes of the given algebraic curve, or asymptotes parallel to the x -axis, are derived by equating the coefficient of the highest power of x in the equation of the supplied curve to zero.

Step 5: Region

Find out the regions of the plane in which no part of the curve lies. To determine such regions, we solve the given equation for y in terms of x or vice-versa. Suppose that y becomes imaginary for $x > a$, the curve does not lie in the region $x > a$.

Step 6: Critical Points

Find out the values of x at which $\frac{dy}{dx} = 0$.

At such points y generally changes its character from an increasing function of x to a decreasing function of x or vice-versa.

Step 7:

Trace the curve with the help of the above points.